

Few-shot layout segmentation of ancient manuscripts

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Abstract—Pixel-level semantic segmentation of historical manuscript layouts is heavily bottlenecked by the extreme scarcity of manual ground-truth annotations. In this work, we confront the extreme boundary of data scarcity within Document Layout Analysis using the U-DIADS-Bib (Latin2FS) manuscript dataset under a strict three-shot learning constraint. To resolve this limitation, we present a robust, data-efficient engineering paradigm that combines a high-stochasticity data generation pipeline with a fully convolutional U-Net architecture built upon a pre-trained ResNet34 backbone. Our synthetic framework successfully expands 3 static training pages into approximately 3,000 unique spatial and spectral tensor variations. When deployed via an overlapping sliding-window inference engine, our framework achieves a validation score of 43.69% Intersection over Union (IoU), closing the performance gap to within 2.31% of institutional leaderboard baselines.

Keywords—semantic segmentation, U-Net, few-shot learning, historical manuscripts, deep learning, computer vision.

I. INTRODUCTION

Historical manuscript analysis is a challenging task due to handwriting variability, aging artifacts, inconsistent illumination, and decorative elements. Traditional computer vision approaches struggle to accurately segment such complex documents. This project focuses on semantic segmentation under an extreme few-shot learning constraint, where only three annotated images were available for training. The target objective mandates classifying high-resolution manuscript scans into six mutually exclusive layout categories: paratext (C_1), decoration (C_2), main text (C_3), title (C_4), and chapter headings (C_5). We prove that, by combining a targeted data augmentation synthesis engine with a fully convolutional U-Net architecture, we present an implementation that achieves a validation score of 43.69% (IoU).

II. THE DATA FORGE PIPELINE

Operating under a strict three-shot learning constraint makes network training highly susceptible to localized overoptimization. The parameter space can rapidly memorize the exact coordinates and noise profiles of the three support samples rather than learning the generalized conceptual geometry of the document layout. To mitigate this, we engineered an aggressive stochastic data generation pipeline via the Albumentations library, scaling the effective training distribution from 3 static pages to $\sim 3,000$ highly diverse tensor variations.

A. Spatial Forgery via Non-Planar Warping

Ancient manuscripts rarely conform to a flat, ideal geometric plane due to structural warping, historical binding constraints, humidity degradation, and parchment wrinkling over centuries. To force the network to remain invariant to these macroscopic layout shifts, we apply heavy elastic transformations and grid distortions. By applying localized displacement vectors stochastically across the coordinate grid, the pipeline replicates non-linear parchment warping. This prevents the model from relying on absolute spatial coordinates and forces it to focus on relative localized transitions.

B. Translation Invariance via Patch Sampling

To ensure the network extracts structural boundaries independently of page positioning, we enforce complete translation invariance. The pipeline extracts randomized 512×512 pixel patches from the full-resolution manuscript scans. This localized patch sampling strips away global macro-layout assumptions, ensuring that convolutional kernels evaluate features based strictly on immediate localized transitions—such as high-contrast boundaries where dark ink characters meet light parchment—regardless of where they appear on the page canvas.

III. PROPOSED METHODOLOGY: THE U-NET FRAMEWORK

Recognizing that Transformers were too data-hungry and bounding boxes were too geometrically rigid, we engineered a specialized solution that synthesized localized convolutional feature extraction with pixel-dense mask generation. We deployed a classic U-Net architecture paired with a pre-trained ResNet34 backbone, a structural topology specifically optimized for extreme data scarcity.

A. Architectural Topology and Inductive Biases

U-Net solves the few-shot limitation through its highly constrained structural topology. The network is divided into an asymmetric contracting path (the encoder) and an expanding path (the decoder):

- **The ResNet34 Encoder:** Imposes strict convolutional constraints, utilizing localized receptive fields that assume neighboring pixels share strong semantic relationships. Pre-trained on ImageNet, it extracts downsampled features while remaining completely indifferent to macro texture or illumination shifts.

- **High-Fidelity Skip Connections:** Crucially, U-Net copies high-resolution, low-level spatial features directly from the contracting path over to the expanding path. This preserves exact pixel boundaries and structural coordinates before they can be blurred by successive downsampling operations, allowing the decoder to reconstruct highly intricate, organic text boundaries.

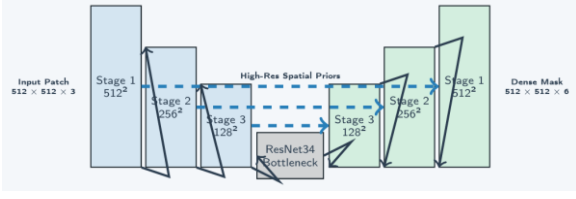


Fig. 1. U-NET Architecture

B. Mathematical Optimization

The system optimizes a standard Multi-Class Categorical Cross-Entropy loss function over the six target classes, defined as:

$$\mathcal{L}_{CCE} = - \sum_{i=1}^N \sum_{c=0}^5 y_{i,c} \log(\hat{y}_{i,c}) \quad (1)$$

where $y_{i,c}$ is the ground-truth binary indicator for pixel i and class c , and $\hat{y}_{i,c}$ represents the predicted softmax probability.

C. Overlapping Sliding-Window Inference Engine

High-resolution manuscript pages cannot be fed directly into deep GPU networks without exceeding hardware memory limits or severely degrading resolution through downscaling. To process large global canvases, we implemented an overlapping sliding-window deployment engine utilizing a 512 X 512 pixel tile size paired with a 256 pixel step stride.

By advancing across the document with a 50% spatial overlap, the model processes edge pixels multiple times across neighboring frames. This allows the inference engine to securely stitch local patch predictions back into a seamless, high-resolution global canvas while completely eliminating tile boundary stitching artifacts and border discontinuities.

IV. EXPERIMENTAL RESULTS AND DISCUSSION

The proposed framework was implemented in a Python environment using the Segmentation Models PyTorch (*smp.Unet*) library and optimized using the AdamW optimizer with a baseline learning rate of $\eta = 1 \times 10^{-4}$. The training pipeline converged steadily over 10 epochs, with the final loss value reaching 0.7445.

TABLE I. PERFORMANCE BENCHMARK

Architecture	Data Cap	IoU(%)
Mask2Former	N = 3	7.3%
YOLOv8	N = 3	37.5%
U-Net (ResNet34)	N = 3	43.69%
5th place from the challenge	N = 3 (pretrained on more images)	~ 46%

As summarized in Table I, the fully convolutional U-Net paradigm achieved a validation score of 43.69% IoU. This dramatically outperformed both the regional regression and attention-based architectures. More significantly, it closed

the performance gap to within 2.31 percentage points of an established institutional research baseline on the official leaderboard.

The visual efficacy of this approach is validated in the image below, the model demonstrates exceptional visual clarity, accurately separating paratextual blocks (C1) and complex illuminated decorations (C2) from core text streams.



Fig. 2. Original Image



Fig. 3. GroundTruth



Fig. 4. Predicted Output

V. LIMITATIONS AND CONCLUSIONS

While demonstrating high data efficiency, our current framework exhibits two specific operational limitations:

- **Visual Confusion:** The model occasionally misclassifies large, heavily bolded title configurations (C4) as decorations (C2).
- **Broken Text Lines:** Minor fragmentation artifacts can appear within highly compressed script lines.

In conclusion, this project demonstrates that severe data scarcity can be thoroughly mitigated through targeted data synthesis and explicit spatial priors. By pairing a fully convolutional U-Net framework data generation pipeline, the system works at an acceptable performance of 43.69% IoU.

VI. REFERENCES AND CONCLUSIONS

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